When should a steel plant stop burning coal?

Investment-timing model, European integrated works · generated 2026-09-22 from the model run, not typed in

2029
commitment year T*, published case

EUR 4.0bn
NPV for a 2.5 Mt/yr plant, published case

EUR 1.3bn
the same plant on sourced inputs

€110
carbon price that justifies it

Read the unit before the number

Every euro figure in this model is **per tonne of annual nameplate capacity** — not per tonne of steel, and not a total. So capex `I0 = €900` is **EUR 2.2bn** for a 2.5 Mt/yr works, and the NPV is in billions. Multiply by annual tonnes and the denominator cancels; that conversion is exact, not indicative.

A carbon price (€/tCO_2) and an operating flow (€/t of product per year) have different denominators and do *not* scale with plant size. Only the capacity-denominated stocks convert.

How much is it worth? T2

CASE	CAPEX I0	HYDROGEN	NPV	NPV, 2.5 MT/YR	TRIGGER	T*
	€/t cap	*€/kg*	*€/t cap*	*€ absolute*	*€/tCO_2*	*year*
base	**900**	**2.60**	**1,606**	**EUR 4.0bn**	**109.5**	**2029**
low	1,200	3.69	991	EUR 2.5bn	172.4	2031
central	**1,420**	**5.00**	**537**	**EUR 1.3bn**	**239.4**	**2038**
high	2,600	8.65	-124	-EUR 310m	460.4	none

Four input sets, from the specification's published values down to ones taken from primary documents. The answer is a **range**, not a figure: EUR 4.0bn down to EUR 1.3bn, and negative at the top of the sourced hydrogen range. Which end you believe is the whole argument.

The comparison that tests it T3

PROJECT	ANNOUNCED	CAPACITY	CAPEX	OUTCOME
	€ total	*Mt/yr*	*€/t cap*	
tkH2Steel Duisburg	EUR 3.0bn	2.3	1,304	BUILDING
SALCOS stage 1	EUR 2.7bn	1.9	1,421	BUILDING
Boden	EUR 6.5bn	2.5	2,600	BUILDING
Bremen + Eisenhuettenstadt	EUR 2.5bn	3.5	714	CANCELLED
Dunkirk EAF	EUR 1.2bn	2	600	CONFIRMED
this model, I0	**EUR 2.2bn**	**2.5**	**900**	**-**

Stegra's Boden is 2.5 Mt/yr and cost €6.5bn to build. The published case says the switch alone is worth EUR 4.0bn on a modelled spend of EUR 2.2bn. Every project still under construction costs at least €1,300/t of capacity; the two below the model's €900 are an EAF-only retrofit and a scheme ArcelorMittal cancelled, forfeiting €1.3bn of approved German aid.

Does it still hold today? T8

Three inputs can be checked against a live primary source — the EU ETS auctions, which publish every clearing price and the money each member state and fund receives, the day it clears (2026-08-24..2026-09-22).

- **Carbon price.** Anchored at €76/tCO_2 for 2026; the auctions cleared **€84.02**. On 4.75 MtCO_2 a year that is **EUR 38m a year** on the plant's carbon bill — waiting is dearer, not cheaper.
- **Volatility — the one that moves the answer.** σ is assumed 0.20; realised is **0.31**, the specification's own 'crisis' rung. There the trigger is €140 not €110, and the path crosses it in **2035** not 2030 — 5 more years of waiting.
- **The pot.** Auctions paid out **EUR 27.5bn** this year; the Innovation Fund, which `s_max` is modelled on, got EUR 2.0bn. The model hands *one* plant EUR 500m — **25%** of that fund, and none of it earmarked for steel.

None of this recalibrates anything: `Params()` still carries the published base case. A year of auctions is not a path to 2050.

What is checked

- **The validation gate.** Nine published quantities reproduce to 1e-3 relative before anything downstream prints.
- **Every table cell re-derived.** 145 figures recomputed from the raw CSV cell or cached statistic — not from the function that prints them — and compared at the precision shown. **145/145 agree.**
- **The specification, quantity by quantity.** 54 of 62 published figures reproduce; all eight differences are NPV *levels*, high by 1.1–8.0%. No year, trigger or crossing year differs.
- 246 tests and 41 standard-library self-checks, run on every build.

What is not

- **Provenance.** Only 9 of 98 rows in the September 2026 research pass carry a working link. The rest name their source correctly without one — *unverified*, not *sourced*. None of it has been promoted into the running model.
- **Operating costs.** The whole BF-BOF against H2-DRI-EAF difference is one constant driven by the hydrogen price. Real lines the model does not contain: DR-grade pellet premium, EAF electricity (€89/t in Germany against €47/t in Sweden), graphite electrodes, scrap and alloys. Closing this needs one study covering both routes at one boundary; none was found.
- **No tax layer.** Electricity is dearer than gas per kWh in every member state, 1.8× to 3.9×, which is why a green hydrogen floor sits above a grey one. Those prices exclude taxes and levies, and the decomposition that "fiscal harmonisation" turns on was not obtained.
- **No MATLAB reference.** The 50-case diff harness is built and its negative control works; there is no reference file to run it against.

The three choices the model will not make

- **Support basis.** A capex lump or a per-tCO_2 stream. The German CCfD pays a stream over fifteen years, worth 1.3–3.3× the €200/t lump the model nets off.
- **Allowance resale.** `w = 1` drives the policy-neutrality result. Without it the headline reverses sign.
- **Whether €2.60/kg hydrogen is the base case at all.** It sits below every green floor and inside the grey band — a fossil price carrying green emissions.

Explanatory framework, not investment advice. Full workings: `results.html` (every table, named and sourced), `charts.html`, `deck.html`. Read `CALIBRATION.md` before quoting any figure.